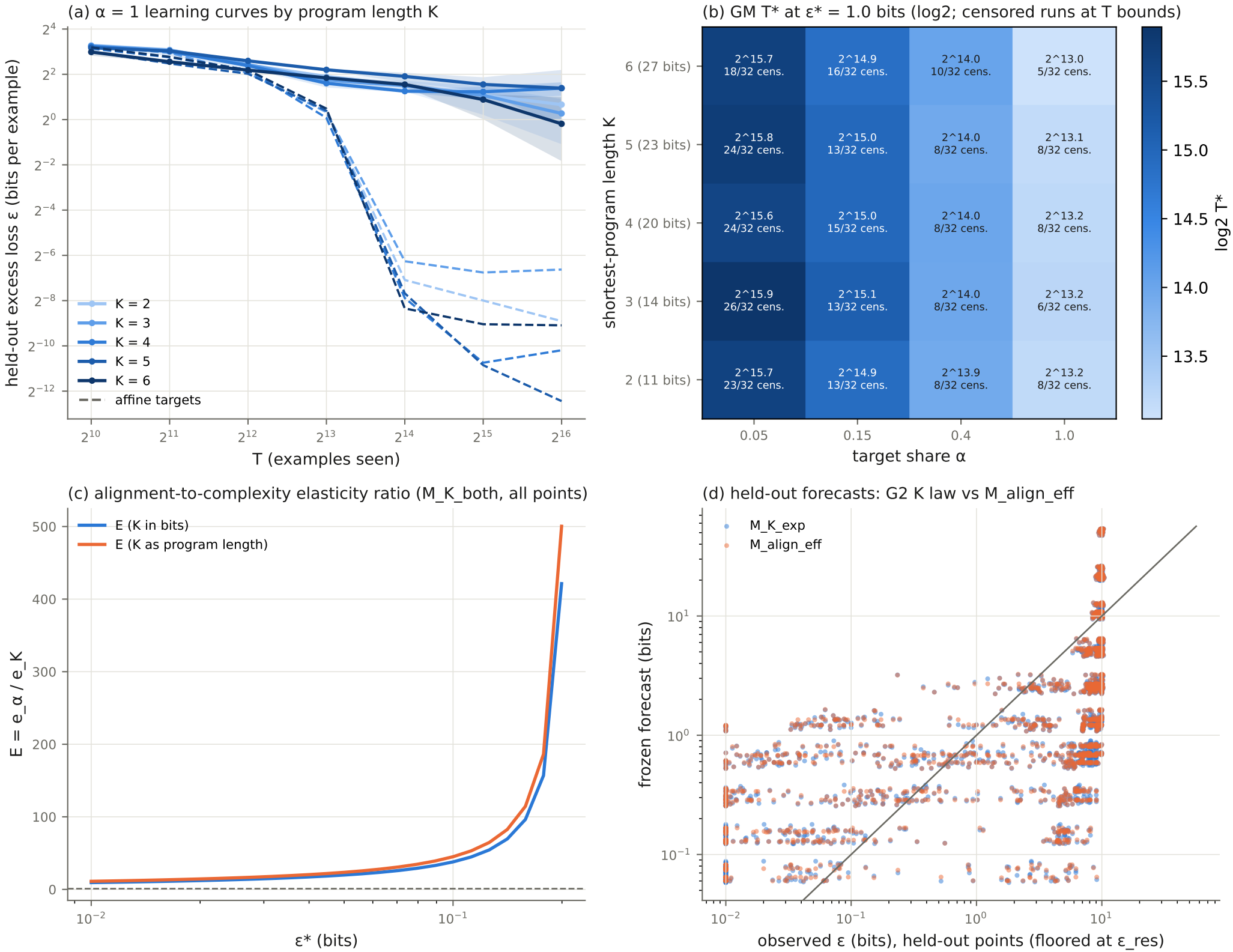

(a) α = 1 learning curves by program length K
held-out excess loss ε (bits per example)
T (examples seen)
K = 2
K = 3
K = 4
K = 5
K = 6
affine targets
(b) GM T* at ε* = 1.0 bits (log2; censored runs at T bounds)
shortest-program length K
6 (27 bits)
5 (23 bits)
4 (20 bits)
3 (14 bits)
2 (11 bits)
2^15.7 18/32 cens.
2^14.9 16/32 cens.
2^14.0 10/32 cens.
2^13.0 5/32 cens.
2^15.8 24/32 cens.
2^15.0 13/32 cens.
2^14.0 8/32 cens.
2^13.1 8/32 cens.
2^15.6 24/32 cens.
2^15.0 15/32 cens.
2^14.0 8/32 cens.
2^13.2 8/32 cens.
2^15.9 26/32 cens.
2^15.1 13/32 cens.
2^14.0 8/32 cens.
2^13.2 6/32 cens.
2^15.7 23/32 cens.
2^14.9 13/32 cens.
2^13.9 8/32 cens.
2^13.2 8/32 cens.
0.05
0.15
0.4
1.0
target share α
log2 T*
(c) alignment-to-complexity elasticity ratio (M_K_both, all points)
E (K in bits)
E (K as program length)
E = e_α / e_K
ε* (bits)
(d) held-out forecasts: G2 K law vs M_align_eff
M_K_exp
M_align_eff
frozen forecast (bits)
observed ε (bits), held-out points (floored at ε_res)